

# Titanic Survival Analysis & Demographic Insights Report

*End-to-End Data Analytics Final Project Report | Problem Statement, Data Cleaning, EDA, Dashboard & Key Insights*

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**Submission:** Github Project Repository

**Dataset Scope:** 891 Passengers (Training Set)

**Company:** SWYNEX Technologies

## 1. Project Overview & Problem Statement

The sinking of the RMS Titanic is one of the most infamous shipwrecks in history. On April 15, 1912, during her maiden voyage, the Titanic sank after colliding with an iceberg, resulting in the death of 1,502 out of 2,224 passengers and crew. While there was an element of chance involved in surviving the disaster, certain groups of people were more likely to survive than others, primarily driven by demographic and socioeconomic factors.

### Core Problem Statement

Analyzed the 891 passenger records from the Titanic dataset to uncover key factors influencing passenger survival, quantify demographic and socioeconomic disparities, and build an interactive visual dashboard alongside actionable behavioral insights.

### Project Objectives

- 1. Data Pipeline Integrity:** Execute structured data profiling, handle missing values (Age, Embarked, Cabin) and ensure schema consistency across variables.
- 2. Exploratory Churn & Survival Drivers:** Evaluate survival variations based on passenger class (Pclass), gender (Sex), age, fare, and family structure (SibSp, Parch).
- 3. Visual Intelligence:** Construct an interactive visual analytics dashboard using Python (Plotly/Seaborn) to highlight survival patterns.
- 4. Key Strategic Insights:** Synthesize operational and safety policy insights, emphasizing priority protocols, socioeconomic equity, and evacuation dynamics.

## 2. Dataset Information & Schema Specification

The primary dataset consists of demographic, travel, and socio-economic information for 891 passengers aboard the RMS Titanic. Below is the full data schema dictionary.

| Column Name  | Data Type | Description                                | Business Scope/Range                                                      | Missing Values |
|--------------|-----------|--------------------------------------------|---------------------------------------------------------------------------|----------------|
| Passenger ID | Integer   | Unique identifier for each passenger       | 1 to 891                                                                  | 0 (0.0%)       |
| Survived     | Binary    | Target variable (0=Perished, 1=Survived)   | 0 or 1                                                                    | 0 (0.0%)       |
| PClass       | Integer   | Socio-economic Ticket Class                | 1=1 <sup>st</sup> class, 2=2 <sup>nd</sup> class, 3=3 <sup>rd</sup> class | 0 (0.0%)       |
| Name         | String    | Full passenger name including formal title | Text string                                                               | 0 (0.0%)       |
| Sex          | String    | Gender of the passenger                    | Male, Female                                                              | 0 (0.0%)       |
| Age          | Float     | Age in years                               | 0.42 to 80 years                                                          | 177 (19.8%)    |
| SibSp        | Integer   | No. of siblings/spouses aboard             | 0 to 8                                                                    | 0 (0.0%)       |
| Parch        | Integer   | No. of parents/children aboard             | 0 to 6                                                                    | 0 (0.0%)       |
| Ticket       | String    | Ticket identification number               | Alphanumeric set                                                          | 0 (0.0%)       |
| Fare         | Float     | Passenger ticket fare paid (£)             | £0.00 to £512.33                                                          | 0 (0.0%)       |
| Cabin        | String    | Cabin number allocated                     | Alphanumeric string                                                       | 687 (77.1%)    |
| Embarked     | String    | Port of Embarkation                        | C = Cherbourg, Q = Queenstown, S = Southampton                            | 2 (0.2%)       |

## 3. Data Cleaning & Transformation Process

To ensure analytical integrity, rigorous data pre-processing and missing value treatments were applied prior to exploratory analysis:

### Step 1: Missing Value Imputation Strategy

- **Age Imputation:** Age missing values (177 missing / 19.8% of records) were imputed using median age grouped by passenger class (Pclass) and title derived from Name (e.g., Master, Miss, Mr, Mrs) to preserve demographic context.
- **Embarked Imputation:** Embarked missing values (2 records) were imputed using the dataset mode ('S' Southampton).
- **Cabin Transformation:** Due to an extreme missingness rate (77.1%), Cabin was converted into a binary flag indicator (Has\_Cabin: 1 if Cabin recorded, 0 otherwise) rather than dropping the field entirely.

### Step 2: Feature Engineering & Duplicate Checks

- **Family Size Construction:** Calculated as `SibSp + Parch + 1` to capture total family presence, alongside a binary indicator `Is Alone` (1 if Family Size = 1).
- **Title Extraction:** Extracted formal titles from the `Name` column into simplified categories (Mr., Mrs., Miss, Master, Rare).
- **Deduplication:** Verified zero duplicate rows across Passenger Id and ticket entries.

## 4. Exploratory Data Analysis & Key Findings

Exploratory analysis revealed stark inequalities in survival distribution based on demographic and socioeconomic groupings:

### Key Survival Breakdown across Passenger Segments

| Segment Category      | Sub-Group/Level                | Total Count | Survived Count | Survival Rate (%) | Analytical Insight/Risk Delta                        |
|-----------------------|--------------------------------|-------------|----------------|-------------------|------------------------------------------------------|
| Overall Dataset       | All Passengers                 | 891         | 342            | 38.4%             | Baseline survival benchmark                          |
| Gender                | Female Passengers              | 314         | 233            | 74.2%             | Highest priority cohort ("Women and children first") |
| Gender                | Male Passengers                | 577         | 109            | 18.9%             | Severe disparity vs females (-55.3% delta)           |
| Ticket Class (Pclass) | 1 <sup>st</sup> class (Upper)  | 216         | 136            | 62.9%             | Upper-deck proximity & evacuation priority           |
| Ticket Class (Pclass) | 2 <sup>nd</sup> class (Middle) | 184         | 87             | 47.3%             | Moderate survival rate                               |
| Ticket Class (Pclass) | 3 <sup>rd</sup> class (Lower)  | 491         | 119            | 24.2%             | Severe overcrowding & lower deck exit delays         |
| Family Structure      | Small Family (2-4)             | 292         | 169            | 57.9%             | Optimal balance of assistance and mobility           |
| Family Structure      | Solo Traveler (1)              | 537         | 163            | 30.4%             | Lack of immediate assistance network                 |
| Family Structure      | Large Family (5+)              | 62          | 10             | 16.1%             | Coordination difficulties in large groups            |
| Embarkation Point     | Cherbourg (C)                  | 168         | 93             | 55.4%             | Higher concentration of 1st class passengers         |
| Embarkation Point     | Queenstown (Q)                 | 77          | 30             | 39.0%             | Mostly 3rd class passengers                          |
| Embarkation Point     | Southampton (S)                | 644         | 217            | 33.7%             | Primary embarkation port                             |

### Detailed Analytical Syntheses

**1. Gender Priority Dominance:** Female passengers survived at a rate of 74.2% compared to only 18.9% for male passengers, confirming strict enforcement of maritime protocol.

**2. Socioeconomic Deck Access Disparity:** 1st Class passengers enjoyed a 62.9% survival rate compared to 24.2% for 3rd Class passengers, reflecting cabin placement advantages closer to the upper lifeboat decks and earlier evacuation notices.

**3. Family Size Dynamics:** Passengers traveling in small families (2-4 members) achieved the highest survival probability (~58%), whereas solo travelers (~30%) and large families (>5 members, ~16%) faced significantly higher mortality.

## 5. Executive Dashboard & Visual Intelligence

An interactive dashboard was created using Python, Plotly, and Seaborn to visualize passenger demographics and survival probabilities:

### Executive Dashboard Architecture

- Panel A (KPI Overview): Total Passengers Analyzed (891), Overall Survival Rate (38.4%), Female Survival (74.2%), Male Survival (18.9%).
- Panel B (Demographic Breakdown): Survival Matrix by Pclass & Sex, Age Distribution Violin Plots (Children vs Adults).
- Panel C (Interactive Filters): Dynamic filtering by Embarked Port, Family Size Category, and Fare Range.

## 6. Key Business & Operational Insights

Lessons derived from the historical Titanic analysis highlight critical considerations for emergency operations, logistics, and resource allocation:

**Universal Accessibility & Socioeconomic Equity:** Clear, unhindered access to emergency infrastructure must be guaranteed across all ticket/pricing tiers, avoiding deck-level isolation during crises.

**Group & Family Evacuation Planning:** Evacuation protocols must account for family unit dynamics to prevent group fragmentation during crisis management.

**Emergency Communication & SLA Response:** Transparent communication and automated alert systems are vital to eliminate delay gaps between upper and lower deck passenger notifications.

## 7. Submission & Repository Documentation

This project has been formatted, documented, and pushed to an open-access GitHub repository for technical verification.

### GitHub Repository Deliverables

- **Repository URL:** [https://github.com/sanjeevani08-tech/swynex\\_project\\_doc](https://github.com/sanjeevani08-tech/swynex_project_doc)
- **Repository Contents:** /data (Raw & Imputed CSVs), /notebooks (EDA & Preprocessing Jupyter Notebooks), /dashboard (Interactive Plotly/Seaborn Scripts & HTML Exports), /reports (Final Project Summary Document).

